KEPLER

User Guide — Installation

Installation

Kepler requires Python 3.8 or later and five supporting libraries (NumPy, Pillow, SciPy, tifffile, scikit-image). The included installer handles everything automatically. Choose the section below for your operating system.

Installing on Windows 11

Kepler runs on Windows 11. The installer checks for Python and all required libraries automatically — if anything is missing it installs it for you. No programming knowledge is needed.

What you need before you start

- A PC running Windows 11 (Home, Pro, or Education).
- An internet connection for the initial installation (to download Python and libraries if needed).
- About 150 MB of free disk space in the worst case — less if Python is already installed.
- Administrator rights are recommended but not required. Right-clicking install.bat and choosing Run as administrator gives the smoothest experience.

Step-by-step installation

1 Extract the Kepler zip file

Right-click the Kepler zip file you downloaded and choose Extract All. Extract it to a convenient location such as your Desktop or Downloads folder. You will see a folder containing install.bat and several other files.

2 Right-click install.bat and run as administrator

Inside the extracted folder, right-click install.bat and choose Run as administrator.

Important:
Windows may show a SmartScreen warning: "Windows protected your PC". This appears because Kepler is newly released, independent software not yet in Microsoft's database. does not mean the software is harmful.

To proceed:

1. Click More info (the small gray link under the warning text)
2. Click Run anyway (a button that now appears)

A terminal window will briefly appear, then the Kepler Installer window will open.

3 Welcome page — click Begin

The Welcome page shows a list of what the installer will do. Click Begin ▶ to continue.

Tip:
If a yellow box says NOT RUNNING AS ADMINISTRATOR, the installer will still work but the system-wide PATH update may be skipped. You can re-run as administrator later if needed.

4 Warnings page — read and click I Understand

This page lists three potential prompts you may see during installation:

UAC prompt ("Allow this app to make changes?"): Click Yes. This appears only if Python needs to be installed.

Python installer checkbox: If the Python installer opens, look for a checkbox at the very bottom of the first screen that says Add python.exe to PATH. Tick it before clicking anything else. This is critical — without it Kepler will not launch from shortcuts or the terminal.

Antivirus prompt: If your antivirus flags pip.exe, choose Allow. This is a known false positive — pip is Python's official package downloader.

Click I Understand ▶ to continue.

5 System Check page — wait for results, then click Continue

The installer scans your system. You will see a checklist of items:

- Green ticks (✓) — already installed, nothing to do.
- Orange items — missing, will be installed automatically.

Wait for all items to resolve. When the Continue ▶ button becomes active, click it.

Tip:

If Python is shown as Not found, the installer will download and install it automatically. You will see a UAC prompt and possibly the Python installer window — follow the instructions in Step 4 above.

6 Installation Options page — review and click Install

Review the installation folder. The default is:

```
C:\Users\YourName\AppData\Local\Kepler
```

You can change this by clicking Browse… if you prefer a different location. Leave the checkboxes ticked:

- Create Desktop shortcut — adds a Kepler icon to your Desktop.
- Add Kepler to the Start Menu — adds Kepler to All Programs.
- Add Kepler to system PATH — lets you type kepler in any terminal.

Click Install ▶ to begin.

7 Installation progress — wait for completion

A progress bar and log show what is happening. The installer will:

- Install or update Python if needed.
- Install NumPy, Pillow, SciPy, tifffile, and scikit-image using pip.
- Copy the Kepler application files to the installation folder.
- Create the Desktop shortcut and Start Menu entry.

Typical time: under 2 minutes if Python is already installed, 5–10 minutes if Python needs to be downloaded.

8 Done — click Finish or Launch Kepler

When the installer shows Kepler Successfully Installed!, you are done.

✓ Kepler has been installed.
You can now launch it from:

- The Kepler shortcut on your Desktop
- Start › All Programs › Kepler
- Any terminal — type: kepler

Tick Launch Kepler when I click Finish if you want to open Kepler immediately, then click Finish.

Troubleshooting — Windows

Problem	Solution
Desktop icon has no image (blank icon)	Run install.bat again as administrator. The icon file was not found at shortcut creation time.
"Python was not found" when double-clicking Kepler	Python's PATH was not set. Re-run the Python installer, tick Add python.exe to PATH, then re-run install.bat.
SmartScreen blocks install.bat	Click More info, then Run anyway. See Step 2 above.
Installer says "pip is not available"	Open a terminal and run: python -m ensurepip --upgrade, then re-run install.bat.
Kepler opens a black window and closes	A library may be missing. Open a terminal and run: pip install numpy Pillow scipy tifffile scikit-image
To uninstall Kepler	Open the installation folder (C:\Users\YourName\AppData\Local\Kepler by default) and run uninstall.bat. This removes Kepler files but leaves Python and its libraries in place.

Installing on Linux Mint

Kepler runs on Linux Mint (and most Ubuntu-based distributions). The installer handles Python library installation automatically using apt and pip, and creates an application menu entry so Kepler appears in your applications like any other program.

What you need before you start

- Linux Mint 20, 21, or 22 (any edition — Cinnamon, MATE, Xfce).
- Python 3.8 or later — this is already included in all recent Linux Mint versions.
- An internet connection for the initial installation.
- About 65 MB of free disk space in the worst case.
- Your user account password (sudo access) — needed only to install the tkinter GUI library if it is missing. Most systems already have it.

Step-by-step installation

1 **Extract the Kepler zip file**

Right-click the Kepler zip file you downloaded and choose Extract Here, or Extract To and pick a location such as your home folder. You will see a folder containing install.sh and several other files.

2 Open a terminal in the Kepler folder

Open the Files app (Nemo), navigate into the extracted Kepler folder, then right-click on an empty area of the folder and choose Open Terminal Here. A terminal window will open, already pointed at the right location.

Tip:
Alternatively, open the Terminal app from the application menu (Menu → Terminal) and then type: cd ~/Downloads/Kepler (or wherever you extracted the files)

3 Run the installer script

In the terminal, type the following command exactly and press Enter:

```
bash install.sh
```

The Kepler Installer window will open after a moment.

Important:
If the terminal says Permission denied, type the following first and then try again:

```
chmod +x install.sh
bash install.sh
```

4 Welcome page — click Begin

The Welcome page shows what the installer will do. It will have already detected your Linux Mint version. Click Begin ▶ to continue.

5 Permissions page — read and click I Understand

This page explains what requires your password and what does not:

Requires your password (sudo): installing python3-tk (the GUI toolkit) and python3-pip if they are missing. The terminal where you ran install.sh will show the password prompt — switch to it if the installer appears to be waiting.

Does not require your password: installing NumPy, Pillow, SciPy, tifffile, and scikit-image (these go into your user account only, using pip --user), copying the Kepler files, and creating the application menu entry.

Click I Understand ▶ to continue.

Tip:
If a green box says sudo is available and cached, the installer will handle everything automatically without pausing for a password.

6 System Check page — wait for results, then click Continue

The installer scans your system. You will see a checklist:

- Green ticks (✓) — already installed.
- Orange items (Missing — will install) — the installer will handle these automatically.

When all items resolve and the Continue ▶ button becomes active, click it.

Important:
If python3-tk shows as Missing, the installer needs to run: sudo apt install python3-tk

You will be prompted for your password in the terminal window. Switch to the terminal, ty your password, and press Enter. Then come back to the installer window.

7 Installation Options page — review and click Install

Review the installation folder. The default is:

```
/home/yourname/.local/share/kepler
```

You can change this by clicking Browse… if you prefer a different location. Leave the checkboxes ticked:

- Create application menu entry — adds Kepler to Menu → Graphics (or similar).
- Create 'kepler' command — lets you type kepler in any terminal.
- Add ~/.local/bin to PATH — ensures the kepler command is always found.

Click Install ▶ to begin.

8 Installation progress — wait for completion

A progress bar and scrolling log show what is happening. The installer will:

- Install any missing system packages via apt (may prompt for your password in the terminal).
- Install NumPy, Pillow, SciPy, tifffile, and scikit-image into your user account using pip --user.
- Copy the Kepler application files to the installation folder.
- Create the application menu entry (.desktop file) and the kepler terminal command.

Typical time: under 2 minutes if all libraries are already installed, 3–5 minutes if pip needs to download them.

9 Done — click Finish or Launch Kepler

When the installer shows Kepler Successfully Installed!, you are done.

✓ Kepler has been installed.
You can now launch it from:

- Menu → Graphics → Kepler (may need to log out and back in once)
- Any terminal — type: kepler
- Direct: python3 ~/.local/share/kepler/main.py

Tick Launch Kepler now and click Finish to open Kepler immediately.

Tip:
If the application menu entry does not appear immediately, log out and log back in. Linux Mint refreshes the application menu on login.

Troubleshooting — Linux Mint

Problem	Solution
"kepler: command not found" in terminal	Run: source ~/.bashrc then try again. Or open a new terminal window. The PATH update takes effect in new sessions.
Installer window does not open (no GUI)	tkinter may be missing. In the terminal run: sudo apt install python3-tk then run: bash install.sh again.
"Permission denied" when running install.sh	Run: chmod +x install.sh then: bash install.sh
Kepler appears in the menu but does not open	Try launching from the terminal: kepler — any error message will be visible there.

Problem	Solution
"No module named 'numpy'" or similar error	Run: pip3 install numpy Pillow scipy tifffile scikit-image --user then try launching Kepler again.
To uninstall Kepler	Open a terminal and run: bash ~/.local/share/kepler/uninstall.sh — this removes the Kepler files, menu entry, and kepler command. Python and its libraries are left in place.